

TAHER Y. AKOLAWALA

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EDUCATION

University of Florida

GPA: 3.73/4.00 — Gainesville, FL

B.S. in Computer Science, Minor in Electrical Engineering

Expected Graduation: May 2028

- **Coursework:** Applied Machine Learning Systems, Data Structures and Algorithms, Operating Systems, Database Systems

EXPERIENCE

ReliaQuest

Tampa, FL

Software Engineering Intern

Jun 2026 – Present

- Built interprocess communication for a legacy C/C++ endpoint agent, implementing subprocess state management and secure interaction with internal data stores (osquery, RocksDB) across 5+ coordinated subprocesses, cutting state-desync failures on agent restart by ~30%.
- Engineered a dynamic config loader on a monitor thread for the endpoint agent's log forwarder, auto-tuning threading, output compression, buffer intervals, and recovery modes from live health/system metrics to cut destination ingress bytes and cost up to ~1.73x while raising throughput EPS ceilings and shrinking device footprint.
- Proposed and designed an AWS load-balancing layer between client log sources (syslog/TCP) and the agent's log forwarders, routing input across agents by live per-agent metrics (EPS throughput, disk-buffer depth, CPU) to prevent buffer bloat and dropped logs during host network disruptions.

Machine Intelligence Lab

Gainesville, FL

Undergraduate Software Engineering Researcher

Jan 2025 – Present

- Refactored and migrated 10,000+ lines of legacy C++ ROS1 code to ROS2, consolidating mission logic into modular Mission classes and reducing per-mission boilerplate by ~35%.
- Developed underwater Gazebo simulations modeling turbidity, buoyancy, and lighting, enabling reproducible testing that cut on-vehicle debugging cycles by ~50%.
- Integrated computer vision frameworks (Detectron2, YOLO) to enhance real-time perception and autonomy, improving object detection accuracy by 25% in simulation.

Summit Analytics

Gainesville, FL

Software Engineering Intern

May 2025 – Aug 2025

- Developed a scalable LinkedIn data scraper using async I/O and BeautifulSoup, automating ingestion into Supabase via REST APIs and improving throughput by 40%.
- Built a natural-language-to-SQL AI agent using Supabase (PostgreSQL) and the OpenAI Agents SDK, enabling real-time querying that cut ad-hoc data requests to engineering by ~50%.
- Created an LLM-driven CSV parser powered by Gemini Flash with TypeScript to normalize and ingest unstructured client data, reducing preprocessing time by 60% and compute cost by 35%.

GatorAI Applied Projects – AI Teaching Assistant

Gainesville, FL

Software Lead

Aug 2025 – Dec 2025

- Led a team of 6 engineers to design and deploy an AI Teaching Assistant, developing RAG, fine-tuning, and guardrail methods used across 3+ courses.
- Architected the end-to-end system integrating LlamaIndex, ChromaDB, and Whisper into a scalable RAG pipeline with Canvas LTI integration, serving responses in under 3 seconds per query.
- Fine-tuned domain-specific LLM models using PEFT/LoRA and implemented Guardrails.ai, reducing off-topic and hallucinated responses by ~40%.

PROJECTS

Drift Zero – Satellite Threat Intelligence Platform | *SCI Hackathon 2026 Winner*

- Led a team of 4 to design and deploy a Databricks ETL pipeline for TLE ingestion and maneuver event detection using SQL window functions, plus operator behavioral profiling to gauge self-clearing likelihood across constellations.
- Engineered 4 adversarial satellite detection modules (incident reconstruction, mission mismatch, resurrection detection, economic impact scoring) using IsolationForest and z-score anomaly detection on orbital element deltas.
- Extended the FastAPI REST backend with 8+ endpoints for real-time anomaly querying and wrote 1,500+ lines of integration tests validating 5 external data sources.

WikiRoute – Wikipedia Graph Pathfinding Engine | *C++, Python, Streamlit*

- Built a scalable C++/Python system modeling Wikipedia's top 100,000 ranked pages as a directed graph, implementing Dijkstra's and Dial's shortest-path algorithms in C++ for high-performance traversal.
- Benchmarked algorithm runtime and memory performance across large-scale graph queries with an interactive Streamlit UI.

Browser4ALL – Voice Browser Automation | *Shell Hacks 2025 Semifinalist*

- Architected a multi-threaded Python system using asyncio with browser-use for browser automation, streaming voice I/O, and multiple API integrations (OpenAI, ElevenLabs).
- Engineered a stateful AI agent with custom context management and 50+ functions for multi-step web navigation and error recovery.

SKILLS

Languages: Python, C, C++, Java, TypeScript/JavaScript, SQL, Bash, PowerShell, HTML/CSS

Frameworks & Libraries: ROS2, PyTorch, TensorFlow, FastAPI, React, Node.js, OpenCV, LlamaIndex, Pandas, NumPy

Developer Tools: Git, GitLab CI/CD, Docker, Linux, Networking (TCP/UDP), WebSockets, osquery, RocksDB, REST APIs, PostgreSQL, MongoDB, Supabase

Platforms & Cloud: AWS, GCP, Databricks, Gazebo, Vercel